

Supplementary Materials

Companion to Paper 3 — BMJ Health & Care Informatics

Supplementary Appendix A: Reproducible Evidence Enrichment and Governance Pipeline

This pipeline is described in full in Supplementary Appendix A of Paper 1. A summary is provided here for context.

The evidence base was generated through a four-stage computational pipeline with cryptographic integrity verification: (1) multi-source evidence retrieval for 39 cases; (2) confidence-weighted triangulation yielding 76 features across 37 cases; (3) canonicalisation into a versioned dataset (v1.0.0); and (4) repository integration (v3.1.0-eee) with manifest-based verification (106 files, SHA-256, 30/30 validation checks). The pipeline is deterministic: re-execution from identical source data produces identical outputs.

Supplementary Appendix A (continued): Enriched Triangulation Evidence

This pipeline is described in full in Supplementary Appendix A of Paper 1. A summary is provided here for context.

A subsequent evidence enrichment and triangulation pipeline processed 39 cases through multi-source evidence retrieval, yielding 37 fully triangulated cases and 2 blocked cases (insufficient admissible evidence). Across the 37 processed cases, triangulated confidence scores ranged from 0.73 to 0.93, with 21 cases achieving STRONG triangulation integrity and 16 classified as FRAGILE. Evidence grades were distributed as follows: 26 IDEAL, 5 ACCEPTABLE, 6 MINIMUM, and 2 INSUFFICIENT. The two blocked cases (H34: Acclarent TruDi Navigation System; H35: Sonio Detect) were retained as monitoring-only entries with no feature values assigned, consistent with the framework's conservative treatment of insufficient evidence. All enrichment artefacts are archived in the Ethical Alpha Audit Core Scientific Record (Zenodo DOI: to be provided upon publication) with SHA-256 hash-chain verification.

Supplementary Appendix A (continued): Full Worked Example — Six-Layer TJS Record (Primary Safety Threshold)

This content is fully specified in the main manuscript and is not duplicated here to avoid redundancy, in accordance with journal supplementary material guidelines.

Supplementary Appendix B: Three-Layer TJS Worked Example and PhysioNet Threshold Sensitivity

Abbreviated Worked Example: Three-Layer TJS Record (Secondary Operational Threshold) — AI-Driven Clinical Alert System

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PhysioNet Threshold Sensitivity

These patterns provide empirical motivation for the tiered TJS requirement that Primary Safety Thresholds receive all six documentation layers: the sensitivity of governance outcomes to small threshold changes at the safety gate indicates that this threshold's numeric value requires explicit calibration evidence (TJS Layer 3) and documented cliff-edge analysis (TJS Layer 4).

Supplementary Appendix C: Extended Scope and Context

This analysis is subject to several limitations. Retrospective evaluation relies on documented failure cases and may not capture unknown failure modes. Simulation-based findings depend on modelled assumptions. Operational feasibility estimates are not derived from prospective deployment. Accordingly, findings should be interpreted within these constraints.

This study contributes to a coordinated set of analyses examining a non-compensatory evaluation approach across complementary domains, including formal logic, simulation behaviour, parameter specification, retrospective validation, and deployment trade-offs.

Estimated execution times (45–90 minutes for primary audit; 10–25 minutes for secondary validation) are consistent with existing governance activities such as multidisciplinary review processes. These estimates are derived from structured task decomposition and should be interpreted as indicative.

In retrospective evaluation, a subset of documented failure cases satisfied conventional performance-based criteria despite exhibiting clinically material failure modes. Under the evaluated non-compensatory approach, these cases did not meet minimum domain-specific requirements, indicating potential limitations of compensatory evaluation frameworks.

All analyses were conducted using deterministic computational procedures with version-controlled inputs and outputs. Where applicable, artefacts were linked via cryptographic hashing to enable independent verification of result provenance. This design allows verification that outputs correspond to specific, fixed inputs without post hoc modification.

Under uniform or random failure distributions, non-compensatory gating may not outperform weighted models, highlighting defined boundary conditions for applicability.

Simulation-informed estimates indicate that Primary Safety Thresholds constitute a minority of total thresholds (<10%), with most decisions operating under a reduced three-layer documentation requirement. Estimated completion times range from 45–90 minutes (Primary) and 10–25 minutes (Secondary).

The framework is consistent with and operationalises the governance intent of the EU AI Act (Articles 9 and 17) and FDA January 2025 guidance on AI-enabled device software.

TJS Field Mapping to JSON/XML Audit Schemas and EHR Integration Specifications

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Supplementary Appendix D: Parameter and Evidence Calibration

The six structural failure mechanisms identified in this paper converge on two fundamental sensitivity dimensions that constitute the empirical research programme for threshold governance. These axes constitute a two-dimensional empirical problem rather than separate lines of inquiry. Interaction is defined operationally as non-additivity of outcome sensitivity under joint perturbation of threshold parameters and evidence confidence distributions. In this context, an interaction effect is present where the combined influence of parameter variation and evidence variation on deployment decisions differs from the sum of their independent effects. Formal characterisation of this interaction is a prespecified objective for future empirical work using factorial simulation designs and variance-based sensitivity analysis (e.g. Sobol interaction indices).

The first is the parameter calibration problem: the gate thresholds, override validity periods, concurrent override limits, and escalation triggers are provisional design defaults whose optimal values are an empirical question. The companion simulation study indicates that the safety gate threshold is the dominant governance lever; small changes in this parameter produce non-linear effects on deployment outcomes, including cliff-edge behaviour where a 0.05-unit shift reverses deployment decisions. The second is the evidence calibration problem: the quality, provenance, and completeness of governance

evidence determine whether gates are satisfied, and these properties vary substantially across cases and institutions. A governance architecture that is insensitive to evidence quality would be undesirable—it would approve poorly documented systems as readily as well-documented ones. The sensitivity of the gate architecture to both parameter calibration and evidence quality is therefore a designed feature, not a limitation, provided that both dimensions are governed transparently. A governance architecture that is insensitive to evidence quality or threshold calibration would be normatively undesirable, as it would produce invariant decisions under epistemically distinct conditions. Sensitivity to both parameterisation and evidence quality is therefore a necessary property of a governance architecture intended to reflect underlying risk and uncertainty.

Supplementary Appendix E: Reserved

This appendix letter is reserved to maintain sequential numbering consistency across the coordinated multi-paper submission bundle.

Supplementary Appendix F: Gaming Resistance Pathways and Assessment

Gaming Resistance (Layer 5): Concrete Pathways and Assessment Three concrete exploitation pathways are particularly relevant for Primary Safety Thresholds. Cohort restriction: a developer restricts the validation cohort to subpopulations where the system performs well. Outcome labelling: the choice of gold-standard outcome definition materially affects performance metrics. Post-hoc threshold selection: the governance threshold is selected after reviewing validation results rather than being pre-specified. For Primary Safety Thresholds, auditor independence is an additional requirement: the party conducting the gaming resistance assessment should have no commercial interest in the outcome.

Supplementary Appendix G: TJS Mapping to Existing NHS Governance Artefacts

Mapping TJS to Existing NHS Governance Artefacts The six TJS layers map to named artefacts already in use across NHS trusts: L1 (Harm-Control Statement) maps to the clinical hazard log maintained under DCB0129 and DCB0160; L2 (Context Key) maps to the intended-use statement required by DCB0129 and by the DTAC transparency criterion; L3 (Calibration Evidence) maps to the NICE Evidence Standards Framework Tier 3b requirements; L4 (Sensitivity Analysis) maps to and extends the clinical risk assessment required by DCB0129; L5 (Gaming Resistance) maps to the independent assessment principle embedded in the MHRA Software and AI as a Medical Device guidance; and L6 (Lifecycle Validity Rule) maps to the post-market surveillance obligations under MHRA guidance and the ongoing monitoring commitments required under the DTAC clinical safety criterion. All mappings are interpretive and illustrative; they demonstrate compatibility with existing governance infrastructure rather than implying that these are the only valid operationalisations.

This work forms part of a coordinated multi-paper research programme examining non-compensatory governance architectures across formal, empirical, and operational domains.